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United States Patent Application Publication	20250273644
Kind Code	A1
Publication Date	August 28, 2025
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SEMICONDUCTOR PACKAGE AND METHOD OF MANUFACTURING THE SEMICONDUCTOR PACKAGE

Abstract

A semiconductor package includes a buffer die; a plurality of core dies sequentially stacked on the buffer die via conductive bumps; and a plurality of adhesive layers between the plurality of core dies, the plurality of adhesive layers filling spaces between the conductive bumps and interconnecting the plurality of core dies; wherein each of the core dies of the plurality of core dies includes a middle region and diagonal regions that extend from the middle region to four corner portions of a respective core die, and wherein the conductive bumps includes: a plurality of first bump structures in the middle region, each first bump structure having a circular shape; and a plurality of second bump structures in each of the diagonal regions, each second bump structure having an elliptical shape.

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Appl. No.: **18/974311**

Filed: **December 09, 2024**

Foreign Application Priority Data

KR	10-2024-0027366	Feb. 26, 2024

Publication Classification

Int. Cl.: **H01L25/18** (20230101); **H01L23/00** (20060101); **H01L25/00** (20060101); **H10B80/00** (20230101)

U.S. Cl.:

CPC **H01L25/18** (20130101); **H01L24/13** (20130101); **H01L24/14** (20130101); **H01L24/16** (20130101); **H01L24/32** (20130101); **H01L24/73** (20130101); H01L24/81 (20130101); H01L25/50 (20130101); H01L2224/13014 (20130101); H01L2224/13147 (20130101); H01L2224/1357 (20130101); H01L2224/14051 (20130101); H01L2224/16145 (20130101); H01L2224/16225 (20130101); H01L2224/32145 (20130101); H01L2224/32225 (20130101); H01L2224/73204 (20130101); H01L2224/81815 (20130101); H01L2225/06513 (20130101); H01L2225/06541 (20130101); H01L2924/1431 (20130101); H01L2924/14361 (20130101); H01L2924/20751 (20130101); H01L2924/37001 (20130101); H10B80/00 (20230201)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2024-0027366, filed on Feb. 26, 2024 in the Korean Intellectual Property Office (KIPO), the contents of which are herein incorporated by reference in their entirety.

BACKGROUND

1. Field

[0002] Example embodiments relate to a semiconductor package and a method of manufacturing the semiconductor package. More particularly, example embodiments relate to a semiconductor package including a plurality of different stacked chips and a method of manufacturing the same.

2. Description of the Related Art

[0003] Recently, electronic devices may include a high bandwidth memory (HBM) or a stacked chip package to provide high performances such as a high capacitance and a high speed operation. In manufacturing the HBM memory devices used in such an electronic device, a plurality of semiconductor chips may be vertically bonded on one another using an adhesive film. In the bonding process of the semiconductor chips, the adhesive film having fluidity may flow from a middle region to an edge region, wherein the adhesive film may be cured to fill spaces between the semiconductor chips. However, after the curing, the adhesive film may not be sufficiently filled in chip corners that are relatively far away, thereby lowering the yield.

SUMMARY

[0004] Example embodiments provide a semiconductor package having conductive bumps that is able to improve bonding quality.

[0005] Example embodiments provide a method of manufacturing the semiconductor package.

[0006] According to an aspect of the disclosure, a semiconductor package, including: a buffer die; a plurality of core dies sequentially stacked on the buffer die via conductive bumps; and a plurality of adhesive layers between the plurality of core dies, the plurality of adhesive layers filling spaces between the conductive bumps and interconnecting the plurality of core dies; wherein each of the core dies of the plurality of core dies comprises a middle region and diagonal regions that extend from the middle region to four corner portions of a respective core die, and wherein the conductive bumps includes: a plurality of first bump structures in the middle region, each first bump structure having a circular shape; and a plurality of second bump structures in each of the diagonal regions, each second bump structure having an elliptical shape.

[0007] According to an aspect of the disclosure, a semiconductor package includes: a first semiconductor chip; a second semiconductor chip; a third semiconductor chip; a fourth semiconductor chip; and a plurality of adhesive layers, wherein the first semiconductor chip, the second semiconductor chip, the third semiconductor chip, and the fourth semiconductor chip are sequentially stacked using conductive bumps, wherein the plurality of adhesive layers are between the first semiconductor chip, the second semiconductor chip, the third semiconductor chip, and the fourth semiconductor chip, the plurality of adhesive layers filling spaces between the conductive bumps and interconnecting the first semiconductor chip, the second semiconductor chip, the third semiconductor chip, and the fourth semiconductor chip, wherein each of the first semiconductor chip, the second semiconductor chip, the third semiconductor chip, and the fourth semiconductor chip comprises (i) a substrate having a first surface and a second surface opposite to the first surface, and (ii) first bonding pads provided on the first surface of the substrate, wherein each of the first semiconductor chip, the second semiconductor chip, the third semiconductor chip, and the fourth semiconductor chip has a middle region and diagonal regions that extend from the middle region to four corner portions of a respective semiconductor chip, and wherein the conductive bumps include: a plurality of first bump structures on the first bonding pads in the middle region, each first bump structure having a circular shape, and a plurality of second bump structures on the first bonding pads in each of the diagonal regions, each second bump structure having an elliptical shape.

[0008] According to an aspect of the disclosure, a semiconductor package includes: a first semiconductor chip comprising (i) a first substrate having a first surface and a second surface opposite the first surface, (ii) a plurality of first through electrodes penetrating the first substrate, and (iii) a plurality of first bonding pads provided on the second surface and electrically connected to the plurality of first through electrodes; a second semiconductor chip stacked on the second surface of the first semiconductor chip, the second semiconductor chip comprising (i) a second substrate having a third surface that faces the second surface and a fourth surface opposite to the third surface, and (ii) a plurality of second bonding pads on the third surface to respectively correspond to the plurality of first bonding pads; a plurality of conductive bumps interposed between the plurality of first bonding pads and the corresponding plurality of second bonding pads; and an adhesive layer between the first semiconductor chip and the second semiconductor chip, the adhesive layer filling spaces between the conductive bumps and interconnecting the first semiconductor chip and the second semiconductor chip, wherein the second substrate of the second semiconductor chip has diagonal regions that extend from a middle region to four corner portions of the second semiconductor chip, wherein the plurality of conductive bumps comprise bump structures on the plurality of second bonding pads in each of the diagonal regions, each bump structure having an elliptical shape, and wherein the bump structures are arranged such that longitudinal axes of the bump structures extend toward the middle region.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0009] Example embodiments will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings. FIGS. **1** to **26** represent non-limiting, example embodiments as described herein.

[0010] FIG. **1** is a cross-sectional view illustrating an electronic device in accordance with example embodiments.

[0011] FIG. **2** is a cross-sectional view illustrating a semiconductor package in accordance with example embodiments.

[0012] FIG. **3** is an enlarged cross-sectional view illustrating portion ‘A’ in FIG. **2**.

[0013] FIG. **4** is an enlarged cross-sectional view illustrating portion ‘B’ in FIG. **2**.

[0014] FIG. **5** is a plan view illustrating conductive bumps on a first semiconductor chip of FIG. **2** in accordance with example embodiments.

[0015] FIG. **6** is an enlarged plan view illustrating portion 'D' in FIG. **5** in accordance with example embodiments.
[0016] FIG. **7** is a perspective view of FIG. **6** in accordance with example embodiments.
[0017] FIGS. **8** to **25** are views illustrating a method of manufacturing a semiconductor package in accordance with example embodiments.
[0018] FIG. **26** is a graph showing a change in flow resistance coefficient according to a ratio of long and short axes of a non-conductive film around an oval conductive bump.

DETAILED DESCRIPTION

[0019] Hereinafter, example embodiments will be explained in detail with reference to the accompanying drawings.
[0020] It is to be understood that both the foregoing general description and the following detailed description are exemplary, and it is to be considered that an additional description of the claimed embodiments is provided. Reference signs are indicated in detail in preferred embodiments of the present embodiments, examples of which are indicated in the reference drawings. Wherever possible, the same reference numbers are used in the description and drawings to refer to the same or like parts.
[0021] It will be understood that, although the terms "first", "second", "third", and so on may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are used to distinguish one element, component, region, layer or section from another element, component, region, layer or section. Thus, a first element, component, region, layer or section described below could be termed a second element, component, region, layer or section, without departing from the spirit and scope of the present disclosure.
[0022] It will be understood that when an element or layer is referred to as being "over," "above," "on," "below," "under," "beneath," "connected to" or "coupled to" another element or layer, it can be directly over, above, on, below, under, beneath, connected or coupled to the other element or layer or intervening elements or layers may be present. In contrast, when an element is referred to as being "directly over," "directly above," "directly on," "directly below," "directly under," "directly beneath," "directly connected to" or "directly coupled to" another element or layer, there are no intervening elements or layers present.
[0023] FIG. **1** is a cross-sectional view illustrating an electronic device in accordance with example embodiments.
[0024] Referring to FIG. **1**, an electronic device **10** may include a substrate **20**, an interposer **30**, a first semiconductor device **40** and a second semiconductor device **50**. In one or more examples, the electronic device **10** may further include first underfill members **44** and **54** and second underfill member **34**, and a heat slug **60**.
[0025] In example embodiments, the electronic device **10** may be a memory module having a stacked chip structure in which a plurality of dies (chips) are stacked. For example, the electronic device **10** may include a semiconductor memory device having a 2.5D chip structure. The electronic device **10** including the 2.5D chip structure memory device may include the interposer **30** for electrically connecting the first and second semiconductor devices (electronic components) **40** and **50** to each other. In one or more examples, a 2.5D chip structure may combine multiple integrated circuit dies in a single package without stacking the integrated circuit dies into a three-dimensional (3D) integrated circuit with through-silicon vias.
[0026] In one or more examples, the first semiconductor device **40** may include a logic semiconductor device, and the second semiconductor device **50** may include a memory device. The logic semiconductor device may be an ASIC as a host such as a CPU, GPU, or SOC. The memory device may include a high bandwidth memory (HBM) device.
[0027] In example embodiments, the substrate **20** may be a substrate having an upper surface and a lower surface facing each other. For example, the substrate **20** may be a printed circuit board

(PCB). The printed circuit board may be a multi-layer circuit board having vias and various circuits therein.

[0028] The interposer **30** may be disposed on the substrate **20**. The interposer **30** may be mounted on the substrate **20** through solder bumps **32**. A planar area of the interposer **30** may be smaller than a planar area of the substrate **20**. When viewed from a plan view, the interposer **30** may be disposed within an area of the substrate **20**. As understood by one of ordinary skill in the art, the plan view may be a view of an object from the top.

[0029] The interposer **30** may be a silicon interposer or a redistribution wiring interposer having a plurality of wirings formed therein. The first semiconductor device **40** and the second semiconductor device **50** may be connected to each other through the wirings in the interposer **30** or electrically connected to the substrate **20** through the solder bumps **32**. The silicon interposer may provide a high-density interconnection between the first and second semiconductor devices **40** and **50**.

[0030] In example embodiments, the first semiconductor device **40** may be disposed on the interposer **30**. The first semiconductor device **40** may be mounted on the interposer **30** by a flip chip bonding method. In one or more examples, the first semiconductor device **40** may be mounted on the interposer **30** such that an active surface on which chip pads are formed faces the interposer **30**. The chip pads of the first semiconductor device **40** may be electrically connected to bonding pads of the interposer **30** by conductive bumps **42**. For example, the conductive bumps may include micro bumps (uBumps). In one or more examples, the flip chip bonding method may include interconnecting semiconductor devices to external circuitry with solder bumps deposited on chip pads. To mount the chip to external circuitry, the chip may be flipped over so that the a top side of the chip faces down and aligned so that the pads of the chip match the pads of the external circuit, where solder may be reflowed to complete the interconnect.

[0031] The second semiconductor device **50** may be disposed on the interposer **30** to be spaced apart from the first semiconductor device **40**. The second semiconductor device **50** may be mounted on the interposer **30** by the flip chip bonding method. In one or more examples, bonding pads of the second semiconductor device **50** may be electrically connected to bonding pads of the interposer **30** by conductive bumps **52**. For example, the conductive bumps **52** may include micro bumps (uBumps).

[0032] Although one first semiconductor device **40** and one second semiconductor device **50** are illustrated to be disposed, it will be appreciated that the embodiments are not limited to these configurations.

[0033] In example embodiments, the first underfill members **44** and **54** may be underfilled between the first semiconductor device **40** and the interposer **30** and between the second semiconductor device **50** and the interposer **30**, respectively. The second underfill member **34** may be underfill between the interposer **30** and the substrate **20**.

[0034] The first and second underfill members may include a material having relatively high fluidity to effectively fill small spaces between the first and second semiconductor devices and the interposer and between the interposer and the substrate. For example, the first and second underfill members may include an adhesive containing an epoxy material. For example, the underfill members may be an epoxy-based material that connects a chip to a board by flowing into the gap between them. The underfill members may provide mechanical reinforcement to solder joints, which can increase the life of the chip. The underfill members may also act as a cushion layer during thermal cycling, protecting the solder joints from damage. The underfill members may also improve dropping performance and thermal cycling performance for components like Flip Chip (FC), Chip-Scale Packages (CSP), and Wafer Level Chip Scale Packages (WLCSP).

[0035] In example embodiments, the heat slug **60** may cover the substrate **20** to thermally contact the first and second semiconductor devices **40** and **50**. Thermal interface materials (TIMs) **62** may be provided on upper surfaces of the first and second semiconductor devices **40** and **50**,

respectively. The heat slug **60** may be disposed to thermally contact the first and second semiconductor devices **40** and **50** via the thermal interface materials **62**.
[0036] External connection pads may be formed on the lower surface of the substrate **20**, and external connection members **22** may be disposed on the external connection pads for electrical connection with external devices. For example, the external connection member **22** may be a solder ball. The electronic device **10** may be mounted on a module board via the solder balls to form a memory module.
[0037] In example embodiments, the second semiconductor device **50** may include a buffer die and a plurality of the memory dies (chips) sequentially stacked on the buffer die. The buffer die and the memory dies may be electrically connected to each other by through silicon vias (TSVs). The through silicon vias may be electrically connected to each other by conductive bumps. The buffer die and the memory die may communicate data signals and control signals through the through silicon vias. In one or more examples, the buffer dies may function to repeat signals/data, and may further function to improve the performance of devices with a higher number of memory dies.
[0038] The electronic device **10** may include the high bandwidth memory (HBM) or a multi-chip package to provide high performance such as high capacity and high speed operation.
[0039] As will be described later, the second semiconductor device **50** may include conductive bumps for signal transmission between the stacked chips and adhesive layers between the stacked chips to fill spaces between the conductive bumps to attach the stacked chips. The conductive bumps may include first bump structures having a circular shape and second bump structures having an elliptical shape when viewed in plan view. The elliptical shapes of the second bump structures may prevent the adhesive layer from being underfilled at the corner of the chip. Thus, the yield in bonding processes of the HBM package may be improved.
[0040] Hereinafter, the HBM package device of FIG. **1** will be described.
[0041] FIG. **2** is a cross-sectional view illustrating a semiconductor package in accordance with example embodiments. FIG. **3** is an enlarged cross-sectional view illustrating portion ‘A’ in FIG. **2** in accordance with example embodiments. FIG. **4** is an enlarged cross-sectional view illustrating portion ‘B’ in FIG. **2** in accordance with example embodiments. FIG. **5** is a plan view illustrating conductive bumps on a first semiconductor chip of FIG. **2** in accordance with example embodiments. FIG. **6** is an enlarged plan view illustrating portion ‘D’ in FIG. **5** in accordance with example embodiments. FIG. **7** is a perspective view of FIG. **6** in accordance with example embodiments. FIG. **2** includes a cross-sectional portion taken along the line C-C′ in FIG. **4** in accordance with example embodiments.
[0042] Referring to FIGS. **2** to **7**, a semiconductor package **50** may include a buffer die **100** and first, second, third and fourth semiconductor chips **200***a*, **200***b*, **200***c* and **200***d*, respectively, as core dies that are sequentially stacked on the buffer die **100**, first, second, third and fourth adhesive layers **300***a*, **300***b*, **300***c* and **300***d*, respectively, interposed between the buffer die **100** and the first, second, third and fourth semiconductor chips **200***a*, **200***b*, **200***c* and **200***d*, and a molding member **400** covering the first, second, third and fourth semiconductor chips **200***a*, **200***b*, **200***c* and **200***d*, respectively, on the buffer die **100**. In one or more examples, the semiconductor package **50** may further include conductive bumps **52** provided on a lower surface of the buffer die **100**.
[0043] In this embodiment, the first, second, third and fourth semiconductor chips **200***a*, **200***b*, **200***c* and **200***d*, respectively, may be substantially the same as or similar to each other. Thus, same or similar reference numerals may be used to refer to the same or like elements and any further repetitive explanation concerning the above elements may be omitted.
[0044] The buffer die **100** and the first, second, third and fourth semiconductor chips **200***a*, **200***b*, **200***c* and **200***d*, respectively, may be stacked on a package substrate such as a printed circuit board (PCB) or an interposer. In one or more embodiments, a semiconductor package as a multi-chip package is illustrated as including five stacked semiconductor chips **100**, **200***a*, **200***b*, **200***c* and **200***d*, respectively. However, the semiconductor package may not be limited thereto, and for

example, the semiconductor package may include 8, 12, 16 or 20 stacked semiconductor chips on the buffer die.

[0045] For example, the semiconductor package **50** may include a high bandwidth memory (HBM) device. The HBM package may include a processor chip and a broadband interface for faster data exchange. The HBM package may have an input/output (TSV I/O) structure including a large number of through silicon via structures to implement a wideband interface. Processor chips that require HBM package support may include a central processing unit (CPU), a graphics processing unit (GPU), a microprocessor, a microcontroller, an application processor (AP), or an application specific integrated circuit (ASIC) chip including a digital signal processing core and an interface for signal exchange.

[0046] The semiconductor package **50** may include the semiconductor chip **100** as a buffer die and the first, second, third and fourth semiconductor chips **200***a*, **200***b*, **200***c* and **200***d*, respectively, as memory dies, sequentially stacked. The first, second, third and fourth semiconductor chips **200***a*, **200***b*, **200***d* and **200***d*, respectively, may be electrically connected to each other by through electrodes such as through silicon vias (TSVs). The memory die may include a memory device, and the buffer die may include a controller controlling the memory device.

[0047] In example embodiments, the buffer die **100** may include a first substrate **110**, a front insulating layer **120** provided on a first surface **112** of the first substrate **110**, first bonding pads **130** provided on the front insulating layer **120**, first through electrodes **140** penetrating the first substrate **110**, a backside insulating layer **170** provided on a second surface **114** of the first substrate **110** opposite to the first surface **112**, and second bonding pads **180** embedded within the backside insulating layer **170**. In one or more examples, the dummy die **100** may further include conductive bumps **52** respectively provided on the first bonding pads **130**. The buffer die **100** may be mounted on the interposer **30** of FIG. **1** via the conductive bumps **52**. For example, the conductive bumps **52** may include solder bumps.

[0048] As illustrated in FIG. **2**, the first substrate **110** may have the first surface **112** and the second surface **114** opposite to the first surface **112**. The first surface may be an active surface, and the second surface may be an inactive surface. An active surface may be an interface that perpendicular to a direction of electron flow. An inactive surface may be an interface that defines a spatial boundary of a semiconductor device. Circuit patterns and cells may be formed on the first surface **112** of the first substrate **110**. For example, the first substrate **110** may be a single crystal silicon substrate. The circuit patterns may include transistors, capacitors, diodes, etc. The circuit patterns may constitute circuit elements. Accordingly, the first semiconductor chip **100** may be a semiconductor device in which a plurality of the circuit elements are formed.

[0049] The front insulating layer **120** may be provided on the first surface **112** of the first substrate **110**, that is, the active surface. The front insulating layer **120** may include a plurality of insulating layers **122**, **124** and upper wirings **123** within the insulating layers. In one or more examples, the first bonding pads **130** may be provided in the outermost insulating layer of the front insulating layer **120**.

[0050] The first through electrodes (through silicon vias, TSVs) **140** may be provided to vertically penetrate the first substrate **110** from the first surface **112** to the second surface **114** of the first substrate **110**. A first end portion of the first through electrode **140** may contact the upper wiring of the front insulating layer. However, as understood by one of ordinary skill in the art, the embodiments are not limited to these configurations. For example, the first through electrode **140** may penetrate the front insulating layer and directly contact the first bonding pad **130**.

[0051] The backside insulating layer **170** may be provided on the second surface **114** of the first substrate **110**, for example, the backside surface. The second bonding pads **180** may be provided in the backside insulating layer **170**. The second bonding pad **180** may be disposed on a surface of the first through electrode **140** exposed from the second surface **114** of the first substrate **110**. Accordingly, the second bonding pad **180** may be electrically connected to the first through electrode **140**.

[0052] In example embodiments, the first semiconductor chip **200***a* may include a substrate **210***a*, a front insulating layer **220***a*, first bonding pads **230***a*, a plurality of through electrodes **240***a*, a backside insulating layer **270***a*, and second bonding pads **280***a*. In one or more examples, the first semiconductor chip **200***a* may further include first conductive bumps CB**1** respectively provided on the first bonding pads **230***a*.

[0053] The substrate **210***a* may have a first surface **212***a* and a second surface **214***a* opposite to the first surface **212***a*. The first surface may be an active surface, and the second surface may be an inactive surface. Circuit elements may be formed on the first surface **212***a* of the substrate **210***a*. The front insulating layer **220***a* as an insulation interlayer may be provided on the first surface **212***a* of the substrate **210***a*, for example, a front surface. The front insulating layer **220***a* may include a plurality of insulating layers **222***a*, **224***a* and upper wirings **223***a* within the insulating layers. In one or more examples, the first bonding pad **230***a* may be provided in the outermost insulating layer of the front insulating layer **220***a*. For example, the front insulating layer **220***a* may include a metal wiring layer **222***a* and a first passivation layer **224***a* sequentially stacked on the substrate **210***a*.

[0054] The through electrodes **240***a* may extend from the first surface **212***a* of the substrate **210***a* to the second surface **214***a*. The through electrode **240***a* may be electrically connected to the first bonding pad **230***a* through the upper wirings **223***a*. The backside insulating layer **270***a* may be provided on the second surface **214***a* of the second substrate **210***a*, e.g., a backside surface. The second bonding pad **280***a* may be provided in the backside insulating layer **270***a*. Accordingly, the first and second bonding pad **230***a* and **280***a* may be electrically connected to each other by the through electrode **240***a*.

[0055] As illustrated in FIG. **5**, the first semiconductor chip **200***a* may have a square shape having four sides and four corner portions C**1**, C**2**, C**3** and C**4** between the sides adjacent to each other. The first semiconductor chip **200***a* may include a middle region MR, diagonal regions DR**1**, DR**2**, DR**3** and DR**4** extending from the middle region MR to the four corner portions C**1**, C**2**, C**3** and C**4** respectively, and peripheral regions PR**1**, PR**2**, PR**3** and PR**4** between the diagonal regions. A length of one side of the first semiconductor chip **200***a* may be within a range of about 8 mm to about 15 mm. A length in a diagonal direction of each diagonal region may be within a range of about 4 mm to about 6.5 mm. As illustrated in FIG. **5**, a center of the middle region MR may coincide with a center of the semiconductor chip **200***a*. The middle region MR may be a square shape having a side with a length corresponding to a width of each of the diagonal regions. In one or more examples, the middle region MR may be a rectangular shape defined by a first length and a second length longer than the first length, where the first length defines a first width of two diagonal regions, and the second length defines a second width of two diagonal regions.

[0056] In example embodiments, the first semiconductor chip **200***a* may be stacked on the buffer die **100** via the first conductive bumps CB**1**. The first conductive bumps CB**1** may include first bump structures **250***a* disposed on the first bonding pads **230***a* in the middle region MR respectively, and second bump structures **260***a* disposed on the first bonding pads **230***a* in the diagonal regions DR**1**, DR**2**, DR**3** and DR**4**, respectively. Additionally, the first conductive bumps CB**1** may include third bump structures **255***a* disposed on the first bonding pads **230***a* in the peripheral regions PR**1**, PR**2**, PR**3** and PR**4**, respectively. When viewed in plan view, the first bump structures **250***a* and the third bump structures **255***a* may have a circular shape, and the second bump structures **260***a* may have an oval shape.

[0057] The first bump structure **250***a* may include a first pillar bump **252***a*, which has a circular shape, and a first solder bump **254***a*, which is formed on the first pillar pump **252***a*. The second bump structure **260***a* may include a second pillar bump **262***a*, which has an elliptical shape, and a second solder bump **264***a*, which is formed on the second pillar bump **262***a*. The third bump structure **255***a* may include a third pillar bump **256***a*, which has a circular shape, and a third solder bump **258***a*, which is formed on the third pillar bump.

[0058] For example, the first, second and third pillar pumps may have a single layer structure. The first, second and third pillar pumps may include a plating pattern layer including copper (Cu). The first, second and third solder bumps may include solder. The first, second and third solder bumps may include, for example, tin (Sn), tin/silver (Sn/Ag), tin/copper (Sn/Cu), tin/indium (Sn/In), tin/silver/copper (Sn/Ag/Cu), etc.

[0059] In one or more examples, the first, second, and third pillar pumps may have a multilayer structure. In one or more examples, the first, second and third pillar pumps may include sequentially stacked first, second and third plating pattern layers. For example, the first and third plating pattern layers may include copper (Cu), and the second plating pattern layer may include nickel (Ni).

[0060] In example embodiments, a first adhesive layer **300***a* may be provided to fill a space between the first conductive bumps CB**1** between the buffer die **100** and the first semiconductor chip **200***a*. For example, the first adhesive layer may include a non-conductive film (NCF).

[0061] For example, the first semiconductor chip **200***a* and the buffer die **100** may be attached to each other by a thermal compression process using the non-conductive film. In one or more examples, the thermal compression process may use heat and force to attach the first semiconductor chip **200***a* to the buffer die **100**. In the thermal compression process, the non-conductive film may be liquefied and have fluidity, may flow between the first semiconductor chip **200***a* and the buffer die **100** and between the first conductive bumps CB**1**, and then may be cured to fill the space between the first conductive bumps CB**1**. A portion of the cured first adhesive layer **300***a* may protrude from a side surface of the first semiconductor chip **200***a*.

[0062] In example embodiments, the second semiconductor chip **200***b* may include a substrate **210***b*, a front insulating **220***b*, first bonding **230***b*, a plurality of through electrodes **240***b*, a backside insulating layer **270***b*, and second bonding pads **280***b*. In one or more examples, the second semiconductor chip **200***b* may further include second conductive bumps CB**2** respectively provided on the first bonding pads **230***b*.

[0063] As illustrated in FIG. **4**, the front insulating layer **220***b* as an insulation interlayer may be provided on a first surface **212***b* of the substrate **210***b*, for example, a front surface. The front insulating layer **220***b* may include a plurality of insulating layers **222***b*, **224***b* and upper wirings **223***b* within the insulating layers. In one or more examples, the first bonding pad **230***b* may be provided in the outermost insulating layer of the front insulating layer **220***b*. For example, the front insulating layer **220***b* may include a metal wiring layer **222***b* and a first passivation layer **224***b* sequentially stacked on the substrate **210***b*.

[0064] The through electrodes **240***b* may extend from the first surface **212***b* of the substrate **210***a* to a second surface **214***b*. The through electrode **240***b* may be electrically connected to the first bonding pad **230***b* through the upper wirings **223***b*. The backside insulating layer **270***b* may be provided on the second surface **214***b* of the substrate **210***b*, that is, a backside surface. The second bonding pad **280***b* may be provided in the backside insulating layer **270***b*. Accordingly, the first and second bonding pad **230***b* and **280***b* may be electrically connected to each other by the through electrode **240***b*.

[0065] In example embodiments, the second semiconductor chip **200***b* may be stacked on the first semiconductor chip **200***a* via the second conductive bumps CB**2**. The second conductive bumps CB**2** may include first bump structures **250***b* disposed on the first bonding pads **230***b* in the middle region MR respectively, and second bump structures **260***b* disposed on the first bonding pads **230***b* in the diagonal regions DR**1**, DR**2**, DR**3** and DR**4**, respectively. Although FIG. **4** only illustrates DR**1**, as understood by one of ordinary skill in the art, the structure illustrated in FIG. **4** may be similarly applied to DR**2**, DR**3**, and DR**4**. In one or more examples, the second conductive bumps CB**2** may include third bump structures disposed on the first bonding pads **230***b* in the peripheral regions PR**1**, PR**2**, PR**3** and PR**4**, respectively. When viewed in plan view, the first bump structures

250*b* and the third bump structures may have a circular shape, and the second bump structures **260***b* may have an oval shape.
[0066] In one or more examples, a second adhesive layer **300***b* may be provided to fill a space between the second conductive bumps CB**2** between the first semiconductor chip **200***a* and the second semiconductor chip **200***b*. For example, the second adhesive layer may include a non-conductive film (NCF). The second semiconductor chip **200***b* and the first semiconductor chip **200***a* may be attached to each other by a thermal compression process using the non-conductive film. In the thermal compression process, the non-conductive film may be liquefied and have fluidity, may flow between the second semiconductor chip **200***b* and the first semiconductor chip **200***a* and between the second conductive bumps CB**2**, and then may be cured to fill the space between the second conductive bumps CB**2**. A portion of the cured second adhesive layer **300***b* may protrude from a side surface of the second semiconductor chip **200***b*.
[0067] In example embodiments, the third semiconductor chip **200***c* may be stacked on the second semiconductor chip **200***b* via third conductive bumps CB**3**. The third conductive bumps CB**3** may include first bump structures **250***c* disposed on first bonding pads **230***c* in the middle region MR respectively, and second bump structures **260***c* disposed on the first bonding pads **230***c* in the diagonal regions DR**1**, DR**2**, DR**3** and DR**4**, respectively. Additionally, the third conductive bumps CB**3** may include third bump structures disposed on the first bonding pads **230***c* in the peripheral regions PR**1**, PR**2**, PR**3** and PR**4**, respectively. When viewed in plan view, the first bump structures **250***c* and the third bump structures may have a circular shape, and the second bump structures **260***c* may have an oval shape.
[0068] In one or more examples, a third adhesive layer **300***c* (FIG. **2**) may be provided to fill a space between the third conductive bumps CB**3** between the second semiconductor chip **200***b* and the third semiconductor chip **200***c*. For example, the third adhesive layer **300***c* may include a non-conductive film (NCF). The third semiconductor chip **200***c* and the second semiconductor chip **200***b* may be attached to each other by a thermal compression process using the non-conductive film.
[0069] In example embodiments, the fourth semiconductor chip **200***d* may be stacked on the third semiconductor chip **200***c* via fourth conductive bumps CB**4**. The fourth conductive bumps CB**4** may include first bump structures **250***d* disposed on first bonding pads **230***d* in the middle region MR respectively, and second bump structures **260***d* disposed on the first bonding pads **230***d* in the diagonal regions DR**1**, DR**2**, DR**3** and DR**4**, respectively. In one or more examples, the fourth conductive bumps CB**4** may include third bump structures disposed on the first bonding pads **230***d* in the peripheral regions PR**1**, PR**2**, PR**3** and PR**4**, respectively. When viewed in plan view, the first bump structures **250***d* and the third bump structures may have a circular shape, and the second bump structures **260***d* may have an oval shape.
[0070] In one or more examples, a fourth adhesive layer **300***d* may be provided to fill a space between the fourth conductive bumps CB**4** between the third semiconductor chip **200***c* and the fourth semiconductor chip **200***d*. For example, the fourth adhesive layer may include a non-conductive film (NCF). The fourth semiconductor chip **200***d* and the third semiconductor chip **200***d* may be attached to each other by a thermal compression process using the non-conductive film.
[0071] As illustrated in FIGS. **5** to **7**, the first bump structures **250***a* having a circular shape may be disposed on the middle region MR of the first semiconductor chip **200***a*, and the second bump structures **260***a* having an oval shape may be disposed on the diagonal regions DR**1**, DR**2**, DR**3**, and DR**4** that extending from the middle region MR to the corner regions C**1**, C**2**, C**3** and C**4**, respectively. The second bump structure **260***a* may be arranged such that the long axis (Xa) of the ellipse faces the middle region MR. A ratio (L/D) of the long axis to the short axis of the second bump structure **260***a* may be at least 1.5. The second bump structure **260***a* may have a major axis length (L) within a range of 15 μm to 40 μm and a minor axis length (D) within a range of 8 μm to 20 μm. The first bump structure **250***a* may have a diameter within a range of 10 μm to 20 μm. In

one or more examples, the major axis length is longer than the minor axis length.
[0072] In the thermal compression process of bonding the first semiconductor chip **200***a* onto the buffer die **100**, when the non-conductive film having fluidity flows through the second bump structures **260***a* along a diagonal direction in the diagonal regions DR**1**, DR**2**, DR**3**, and DR**4** that have relatively long lengths, since the flow resistance coefficient CD of the second bump structure **260***a* is relatively low compared to the first and third bump structures, a flow rate of the non-conductive film flowing through the diagonal region may be increased. Accordingly, the time for the non-conductive film to reach from the middle region MR to the corner portions C**1**, C**2**, C**3**, and C**4** may be advantageously reduced, to thereby prevent the non-conductive film from being unfilled in the corner portions.
[0073] As mentioned above, the semiconductor package **50** may include the first, second, third and fourth semiconductor chips **200***a*, **200***b*, **200***c*, and **200***d* sequentially stacked on the buffer die **100** via the first, second, third and fourth conductive bumps CB**1**, CB**2**, CB**3** and CB**4**, and the first, second, third and fourth adhesive layers **300***a*, **300***b*, **300***c*, and **300***d* filling spaces between the first to fourth conductive bumps between the first to fourth semiconductor chips and adhering the first to fourth semiconductor chips to each other. The first, second, third, and fourth conductive bumps may include the first bump structures **250***a*, **250***b*, **250***c*, **250***d* disposed in the middle regions MR of the first to fourth semiconductor chips respectively, and having a circular shape, and the second bump structures **260***a*, **260***b*, **260***c*, **260***d* disposed in the diagonal regions respectively and having an oval shape.
[0074] The second bump structure may be arranged such that the long axis Xa of the oval shape extends towards the middle region MR. A ratio (L/D) of the long axis to the short axis of the second bump structure may be at least 1.5. In the bonding processes of the first to fourth semiconductor chips, the flow resistance coefficient (CD) of the second bump structure with respect to the non-conductive film having fluidity may be relatively reduced compared to the first bump structure due to the oval shape.
[0075] Accordingly, in the bonding process, when the non-conductive film passes through the second bump structures along the diagonal direction in the diagonal regions DR**1**, DR**2**, DR**3**, and DR**4**, the flow rate of the non-conductive film may be advantageously increased. Thus, the time for the non-conductive film to reach from the middle region MR to the corner portions C**1**, C**2**, C**3**, and C**4** may be reduced, thereby preventing the non-conductive film from being unfilled in the corner portions.
[0076] Hereinafter, a method of manufacturing the semiconductor package of FIG. **1** will be described.
[0077] FIGS. **8** to **25** are views illustrating a method of manufacturing a semiconductor package in accordance with example embodiments. FIG. **8** is a plan view illustrating a semiconductor wafer in which first semiconductor chips are formed in accordance with example embodiments. FIGS. **12** and **14** are enlarged cross-sectional views illustrating portion 'G' in FIG. **11** in accordance with example embodiments. FIG. **13** is a plan view illustrating a photoresist pattern formed on the semiconductor wafer of FIG. **12** in accordance with example embodiments. FIG. **14** is an enlarged cross-sectional view illustrating portion 'H' in FIG. **13** in accordance with example embodiments. FIG. **17** is an enlarged cross-sectional view illustrating portion 'I' in FIG. **16** in accordance with example embodiments. FIG. **21** is an enlarged cross-sectional view illustrating portion 'J' in FIG. **20** in accordance with example embodiments. FIG. **23** is an enlarged cross-sectional view illustrating portion 'K' in FIG. **22** Q301010.
[0078] Referring to FIGS. **8** to **10**, a second wafer W**2** in which a plurality of first semiconductor chips (core dies) are formed may be provided.
[0079] In example embodiments, the second wafer W**2** may include a substrate **210***a* having a first surface **212***a* and a second surface **214***a* opposite to the first surface **212***a*. The second wafer W**2** may include a die region DA and a scribe lane region SA surrounding the die region DA. The

second wafer W**2** may be cut along the scribe lane region SA that divides a plurality of the die regions DA of the second wafer W**2** by a following sawing process to be individualized into a plurality of first semiconductor chips.

[0080] The die region DA may have a square shape having four sides and four corner portions between the sides adjacent to each other. The die region DA may include a middle region MR and diagonal regions DR**1**, DR**2**, DR**3**, and DR**4** extending from the middle region MR to the four corner portions, respectively.

[0081] Circuit elements may be formed in the die region DA on the first surface **212***a* of the substrate **210***a*. The circuit element may include a plurality of memory devices. Examples of the memory device may include a volatile semiconductor memory device and a non-volatile semiconductor memory device. Examples of the volatile semiconductor memory device may be DRAM, SRAM, etc. Examples of the non-volatile semiconductor memory device may be EPROM, EEPROM, Flash EEPROM, etc. However, the memory device may be any memory structure known to one of ordinary skill in the art.

[0082] For example, the substrate **210***a* may include silicon, germanium, silicon-germanium, or III-V compounds, e.g., GaP, GaAs, GaSb, etc. In some embodiments, the substrate **210***a* may be a silicon-on-insulator (SOI) substrate, or a germanium-on-insulator (GOI) substrate.

[0083] The circuit elements may include, for example, transistors, capacitors, wiring structures, etc. The circuit elements may be formed on the first surface **212** of the substrate **210***a* by performing a fab process called a front-end-of-line (FEOL) process for manufacturing semiconductor devices. A surface of the second substrate on which the FEOL process is performed may be referred to as a front surface of the substrate, and a surface opposite to the front surface may be referred to as a backside surface. The FEOL process may be the first stage of integrated circuit fabrication, where individual components may be patterned into a semiconductor substrate. The FEOL process may be used to build transistor and other components (e.g., capacitors and resistors) directly inside the wafer. An insulation interlayer may be formed on the first surface **212***a* of the substrate **210***a* to cover the circuit elements.

[0084] In example embodiments, the second wafer W**2** may include a front insulating layer **220***a* provided on the first surface **212***a* of the substrate **210***a*. The front insulating layer **220***a* may include a metal wiring layer **222***a* and a first passivation layer **224***a* sequentially stacked on the substrate **210***a*. The front insulating layer may be formed by performing a wiring process called a back-end-of-line (BEOL) process. The BEOL process may include depositing metal wiring between individual devices on a wafer and may be performed after the FEOL process. The BEOL process may involve electrically connecting transistors and components. A first bonding pad **230***a* may be provided in the outermost insulating layer of the front insulating layer **220***a*.

[0085] The metal wiring layer **222***a* may include a plurality of insulating layers, upper wirings **223***a* within the plurality of insulating layers, and the fist bonding pad **230***a* may be formed on uppermost wirings of the plurality of upper wirings **223***a*. For example, the upper wirings may include aluminum (Al), copper (Cu), tin (Sn), nickel (Ni), gold (Au), platinum (Pt), or an alloy thereof. The first passivation layer **224***a* may be formed on the metal wiring layer **222***a* and may cover at least a portion of the first bonding pad **230***a*. For example, the insulating layers may be formed to include an oxide such as silicon oxide, carbon-doped oxide, or fluorine-doped oxide. The first passivation layer may include a first protective layer including an oxide layer and a second protective layer including a nitride layer, which are sequentially stacked. The first protective layer may include silicon oxide, and the second protective layer may include silicon nitride or silicon carbonitride.

[0086] The first bonding pad **230***a* may be provided in the first passivation layer **224***a*. The first bonding pad **230***a* may be exposed through the outer surface of the first passivation layer **224***a*. In one or more examples, an insulating interlayer may be provided on the first surface **212***a* of the substrate **210***a* to cover the circuit patterns. The insulating interlayer may be formed to include, for example, silicon oxide or a low dielectric material. The insulating interlayer may include lower wirings that are electrically connected to the circuit patterns therein. Accordingly, the circuit pattern may be electrically connected to the first bonding pad **230***a* through the lower wirings and the upper wirings.

[0087] In example embodiments, the second wafer W**2** may include through electrodes **240***a* that at least partially penetrate the substrate **210**. The through electrodes **240***a* may extend from the first surface **212***a* of the substrate **210***a* to a predetermined depth. The through electrode **240***a* may contact the lowermost wiring of the upper wirings. Accordingly, the through electrode **240***a* may be electrically connected to the first bonding pad **230***a* through the upper wirings **223***a*.

[0088] Referring to FIGS. **11** to **18**, first conductive bumps CB**1** as conductive connection members may be formed on the first bonding pads **230***a* on the first surface **212***a* of the substrate **210***a*, respectively.

[0089] First, as illustrated in FIGS. **11** to **14**, a seed layer may be formed on the first bonding pads **230***a* of the substrate **210***a*, and a photoresist pattern PR having openings OP that expose conductive bump regions may be formed on the seed layer. The seed layer may be a thin layer of metal deposited on the first bonding pads **230***a* to improve the properties the following components deposited thereon.

[0090] The seed layer may include titanium/copper (Ti/Cu), nickel/gold (Ni/Au), titanium/palladium (Ti/Pd), titanium/nickel (Ti/Ni), chrome/copper (Cr/Cu) or an alloy thereof. The seed layer may be formed by a sputtering process.

[0091] After a photoresist layer is formed on the seed layer on the first surface **212***a* of the substrate **210***a*, an exposure process may be performed on the photoresist layer to form the photoresist pattern PR having the openings OP. The openings OP of the photoresist pattern PR may include first openings OP**1** that expose first bump structure regions in the middle region MR and second openings OP**2** that expose second bump structure regions in the diagonal regions DR**1**, DR**2**, DR**3**, and DR**4**. The openings OP of the photoresist pattern PR may include third openings OP**3** that expose third bump structure regions in peripheral regions PR**1**, PR**2**, PR**3**, and PR**4** between the diagonal regions.

[0092] When viewed in plan view, each of the first openings OP**1** may have a circular shape and each of the second openings OP**2** may have an elliptical shape having a long axis (major axis) and a short axis (minor axis). The third openings OP**3** may have a shape the same as the first opening OP**1**. The second opening OP**2** may have an oval shape. The second openings OP**2** may be formed such that the long axes Xa of the second openings OP**2** face the middle region MR. The first opening OP**1** may have a diameter W within a range of 10 μm to 20 μm. The major axis of the second opening OP**2** may have a length L within a range of 15 μm to 40 μm, and the minor axis of the second opening OP**2** may a length D within a range of 8 μm to 20 μm.

[0093] As illustrated in FIG. **15**, the first, second and third openings OP**1**, OP**2** and OP**3** of the photoresist pattern PR may be filled up with a conductive material to form the first conductive bumps.

[0094] The first conductive bumps may be formed by performing a plating process on the seed layer. For example, a first bump structure **250***a* having a circular shape may be formed in the first opening OP**1** that is formed in the middle region MR, second bump structures **260***a* having elliptic shapes may be formed in the second openings OP**2** that are formed in the diagonal regions DR**1**, DR**2**, DR**3** and DR**4**, and third bump structure having circular shapes may be formed in the third openings OP**3** that are formed in the peripheral regions PR**1**, PR**2**, PR**3** and PR**4**.

[0095] The first bump structure **250***a* may include a first pillar bump **252***a*, which has a circular shape, and a first solder bump **254***a*, which is formed on the first pillar pump **252***a*. The second bump structure **260***a* may include a second pillar bump **262***a*, which has an elliptical shape, and a second solder bump **264***a*, which is formed on the second pillar bump **262***a*. The third bump structure may include a third pillar bump, which has a circular shape, and a third solder bump, which is formed on the third pillar bump.

[0096] For example, the first, second and third pillar pumps may have a single layer structure. The first, second and third pillar pumps may include a plating pattern layer including copper (Cu). The first, second and third solder bumps may include solder. The first, second and third solder bumps may include, for example, tin (Sn), tin/silver (Sn/Ag), tin/copper (Sn/Cu), tin/indium (Sn/In), tin/silver/copper (Sn/Ag/Cu), etc.

[0097] In one or more examples, the first, second and third pillar pumps may have a multilayer structure. In one or more examples, the first, second and third pillar pumps may include sequentially stacked first, second and third plating pattern layers. For example, the first and third plating pattern layers may include copper (Cu), and the second plating pattern layer may include nickel (Ni).

[0098] Then, the photoresist pattern PR may be removed from the wafer W**2** and portions of the seed layer exposed by the bump structures may be removed to form a seed layer pattern.

[0099] The first conductive bumps CB**1** may include the first bump structures **250***a* disposed on the first bonding pads **230***a* in the middle region MR respectively, and the second bump structures **260***a* disposed on the first bonding pads **230***a* in the diagonal regions DR**1**, DR**2**, DR**3** and DR**4**, respectively. In one or more examples, the first conductive bumps may include the third bump structures disposed on the first bonding pads **230***a* in the peripheral regions PR**1**, PR**2**, PR**3** and PR**4**, respectively. The first bump structure **250***a* may have a first thickness H**1**, and the second bump structure **260***a* may have a second thickness H**2** that is greater than or equal to the first thickness H**1**. The third bump structure may have the first thickness H**1**. The first and second thicknesses H**1** and H**2** may be within a range of 10 μm to 50 μm.

[0100] When the thicknesses of the first and second pillar bumps **252***a* and **262** are relatively small, the first and second pillar bumps **252***a* and **262***a* may be referred to as first bonding pads and the first bonding pads **230***a* may be referred to as first redistribution pads.

[0101] Referring to FIG. **18**, a backside insulating layer **270***a* with a second bonding pad **280***a* may be formed on the second surface **214***a* of the substrate **210***a*.

[0102] First, the structure of FIG. **16** may be turned over, the second wafer W**2** may be attached to a first carrier substrate C**1** using a first adhesive film PF**1**, and then, the second surface **214***a* of the substrate **210***a* may be removed until portions of the through electrodes **240***a* are exposed.

[0103] The second surface **214***a* of the substrate **210***a* may be partially removed by a grinding process, such as a chemical mechanical polishing (CMP) process. Accordingly, a thickness of the substrate **210***a* may be reduced to a desired thickness. For example, the substrate **210***a* may have the thickness within a range of about 30 μm to 150 μm. Then, a silicon recess etching process may be performed on the second surface **214***a* of the substrate **210***a* to expose one end portion of the through electrode **240***a* from the second surface **214***a* of the substrate **210***a*.

[0104] The backside insulating layer **270***a* as a second passivation layer having the second bonding pad **280***a* that is electrically connected to the through electrode **240***a* may be formed on the second surface **214***a* of the substrate **210***a*. For example, after forming the backside insulating layer **270***a* on the second surface **214***a* of the substrate **210***a*, an opening may be formed in the backside insulating layer **270***a* to expose the through electrode **240***a*, and a plating process may be performed to form the second bonding pad **280***a* in the opening. The second bonding pad **280***a* may be disposed on the exposed surface of the through electrode **240***a*. The backside insulating layer **270***a* may include silicon oxide, carbon-doped silicon oxide, silicon carbonitride (SiCN), etc. Accordingly, the first and second bonding pads **230***a* and **280***a* may be electrically connected to each other by the through electrode **240***a*.

[0105] Referring to FIG. **19**, a first adhesive layer **300***a* may be formed on the front insulating layer **220***a* on the first surface **212***a* of the substrate **210***a*, and the second wafer W**2** may be cut along the scribe lane region SA to form an individualized first semiconductor chip **200***a*. The second wafer W**2** may be cut by a sawing process.

[0106] The first adhesive layer **300***a* may be formed to cover the first conductive bumps. For example, the first adhesive layer **300***a* may include a thermosetting resin. The first adhesive layer **300***a* may include a non-conductive film (NCF).
[0107] Referring to FIGS. **20** and **21**, the first semiconductor chip **200***a* may be stacked on a first wafer W**1**. The first semiconductor chip **200***a* may be stacked on the first wafer W**1** via the first conductive bumps. The first semiconductor chip **200***a* may be attached to the first wafer W**1** by the first adhesive layer **300***a*.
[0108] In example embodiments, first, conductive bumps **52** may be formed on a front surface of the first wafer W**1** in which a plurality of buffer dies are formed, and a second adhesive film PF**2** may formed on the front surface of the first wafer W**1** to cover the conductive bumps **52**. Then, the first wafer W**1** may be attached onto a second carrier substrate C**2** using the second adhesive film PF**2**.
[0109] The first wafer W**1** may include a first substrate **110** having a first surface **112** and a second surface **114** opposite to the first surface **112**. The first wafer W**1** may include a die region DA and a scribe lane region SA surrounding the die region DA. The first wafer W**1** may be cut along the scribe lane region SA that divides a plurality of the die regions DA of the first wafer W**1** by a following sawing process to be individualized into a plurality of buffer dies.
[0110] Then, the first semiconductor chip **200***a* may be stacked on the first wafer W**1** by using a substrate support system WSS. The first semiconductor chips **200***a* may be disposed on the first wafer W**1** to correspond to the die regions DA, respectively. The first semiconductor chip **200***a* may be arranged such that the first surface **212***a* of the substrate **210***a* of the first semiconductor chip **200***a* faces the first wafer W**1**.
[0111] The first semiconductor chip **200***a* may be attached on the first wafer W**1** by performing a thermal compression process at a predetermined temperature (e.g., about 400° C. or less). Through the thermal compression process, the first semiconductor chip **200***a* and the first wafer W**1** may be bonded to each other. In one or more examples, the predetermined temperature may be applied for a period of time sufficient to attach the first semiconductor chip **200***a* to the first wafer W**1** and to liquefy the non-conductive film.
[0112] The first conductive bumps of the first semiconductor chip **200***a* may be bonded to second bonding pads **180** of the buffer die through the thermal compression process. The first bonding pad **230***a* of the first semiconductor chip **200***a* may be electrically connected to the second bonding pad **180** of the buffer die through the first conductive bump.
[0113] In the thermal compression process, the non-conductive film may be liquefied and may have fluidity, and may flow between the first semiconductor chip **200***a* and the first wafer W**1**. The non-conductive film having fluidity may flow between the first conductive bumps, and then may be cured to fill a space between the first conductive bumps. A portion of the cured first adhesive layer **300***a* may protrude from a side surface of the first semiconductor chip **200***a*.
[0114] The first bump structures **250***a* having a circular shape may be disposed in the middle region MR of the first semiconductor chip **200***a*, and the second bump structures **260***a* having an oval shape may be disposed in the diagonal regions DR**1**, DR**2**, DR**3**, and DR**4** that extend from the middle region MR of the first semiconductor chip **200***a* to the four corner portions C**1**, C**2**, C**3**, and C**4**, respectively. The third bump structures having a circular shape may be disposed in the peripheral regions of the first semiconductor chip **200***a*. The second bump structure **260***a* may be arranged such that the long axis Xa of the oval shape extends towards the middle region MR. A ratio (L/D) of the long axis to the short axis of the second bump structure **260***a* may be at least 1.5.
[0115] When the non-conductive film having fluidity flows through the second bump structures **260***a* along a diagonal direction in the diagonal regions DR**1**, DR**2**, DR**3**, and DR**4** that have relatively long lengths, since the flow resistance coefficient CD of the second bump structure **260***a* is relatively low compared to the first bump structure, a flow rate of the non-conductive film flowing through the diagonal region may be relatively increased. Accordingly, the time for the non-

conductive film to reach from the middle region MR to the corner portions C**1**, C**2**, C**3**, and C**4** may be reduced, thereby preventing the non-conductive film from being unfilled in the corner portions.
[0116] Referring to FIGS. **22** and **23**, a second semiconductor chip **200***b* may be stacked on the first semiconductor chip **200***a*.
[0117] First, processes the same as or similar to the processes described with reference to FIGS. **8** to **19** may be performed to form an individualized second semiconductor chip **200***b*, and processes the same as or similar to the processes described with reference to FIGS. **20** and **21** may be performed to stack the second semiconductor chip **200***b* on the first semiconductor chip **200***a*.
[0118] In example embodiments, the second semiconductor chip **200***b* may be stacked on the first semiconductor chip **200***a* via second conductive bumps. A second adhesive layer **300***b* may be adhered on the second semiconductor chip **200***b* to adhere the second semiconductor chip **200***b* onto the first semiconductor chip **200***a*. The second adhesive layer **300***b* may be adhered on a second front insulation layer **220***b* to cover the second conductive bumps. The second semiconductor chip **200***b* may be attached onto the first semiconductor chip **200***a* using the second adhesive layer **300***b*. The second semiconductor chip **200***b* may be arranged such that a first surface **212***b* of a substrate **210***b* of the second semiconductor chip **200***b* faces the first semiconductor chip **200***a*. The second adhesive layer **300***b* may include a non-conductive film (NCF).
[0119] The second semiconductor chip **200***b* may be attached on the first semiconductor chip **200***a* by performing a thermal compression process. The second conductive bumps of the second semiconductor chip **200***b* may be bonded to the second bonding pads **280***a* of the first semiconductor chip **200***a* through the thermal compression process. A first bonding pad **230***b* of the second semiconductor chip **200***b* may be electrically connected to the second bonding pad **280***a* of the first semiconductor chip **200***a* through the second conductive bump.
[0120] In the thermal compression process, the non-conductive film may be liquefied and may have fluidity, and may flow between the second semiconductor chip **200***b* and the first semiconductor chip **200***a*. The non-conductive film having fluidity may flow between the second conductive bumps and then may be cured to fill a space between the second conductive bumps. A portion of the cured second adhesive layer **300***b* may protrude from a side surface of the second semiconductor chip **200***b*.
[0121] First bump structures **250***b* having a circular shape may be disposed in a middle region MR of the second semiconductor chip **200***b*, and second bump structures **260***b* having an oval shape may be disposed in diagonal regions DR**1**, DR**2**, DR**3**, and DR**4** that extend from the middle region MR of the second semiconductor chip **200***b* to four corner portions, respectively. Third bump structures having a circular shape may be disposed in peripheral regions of the second semiconductor chip **200***b*. The second bump structure **260***b* may be arranged such that the long axis Xa of the oval shape extends towards the middle region MR. A ratio (L/D) of the long axis to the short axis of the second bump structure **260***b* may be at least 1.5.
[0122] When the non-conductive film having fluidity flows through the second bump structures **260***b* along a diagonal direction in the diagonal regions DR**1**, DR**2**, DR**3**, and DR**4** that have relatively long lengths, since the flow resistance coefficient CD of the second bump structure **260***b* is relatively low compared to the first bump structure, a flow rate of the non-conductive film flowing through the diagonal region may be relatively increased. Accordingly, the time for the non-conductive film to reach from the middle region MR to the corner portions may be reduced, to thereby prevent the non-conductive film from being unfilled in the corner portions.
[0123] Referring to FIG. **24**, processes the same as or similar to the processes described with reference to FIGS. **22** and **23** may be performed to stack a third semiconductor chip **200***c* on the second semiconductor chip **200***b* and stack a fourth semiconductor chip **200***d* on the third semiconductor chip **200***c*.
[0124] In example embodiments, the third semiconductor chip **200***c* may be attached onto the

second semiconductor chip **200**_b_ using a third adhesive layer **300**_c_. The third semiconductor chip **200**_c_ may be arranged such that a first surface **212**_c_ of a substrate **210**_c_ of the third semiconductor chip **200**_c_ faces the second semiconductor chip **200**_b_. The third semiconductor chip **200**_c_ may be stacked on the second semiconductor chip **200**_b_ via third conductive bumps. The third semiconductor chip **200**_c_ may be attached on the second semiconductor chip **200**_b_ by performing a thermal compression process. The third conductive bumps of the third semiconductor chip **200**_c_ may be bonded to second bonding pads **280**_b_ of the second semiconductor chip **200**_b_ through the thermal compression process. A first bonding pad **230**_c_ of the third semiconductor chip **200**_c_ may be electrically connected to the second bonding pad **280**_b_ of the second semiconductor chip **200**_b_ through the third conductive bump. The third adhesive layer **300**_c_ may include a non-conductive film (NCF).

[0125] In the thermal compression process, the non-conductive film may be liquefied and may have fluidity, and may flow between the third semiconductor chip **200**_c_ and the second semiconductor chip **200**_b_. The non-conductive film having fluidity may flow between the third conductive bumps and then may be cured to fill a space between the third conductive bumps. A portion of the cured third adhesive layer **300**_c_ may protrude from a side surface of the third semiconductor chip **200**_c_.

[0126] Second bump structures **260**_c_ having an oval shape may be disposed in diagonal regions that extend from a middle region MR of the third semiconductor chip **200**_c_ to four corner portions, respectively. The second bump structure **260**_c_ may be arranged such that the long axis Xa of the oval shape extends towards the middle region MR. A ratio (L/D) of the long axis to the short axis of the second bump structure **260**_c_ may be at least 1.5.

[0127] When the non-conductive film having fluidity flows through the second bump structures **260**_c_ along a diagonal direction in the diagonal regions that have relatively long lengths, since the flow resistance coefficient CD of the second bump structure **260**_c_ is relatively low compared to the first bump structure, a flow rate of the non-conductive film flowing through the diagonal region may be relatively increased. Accordingly, the time for the non-conductive film to reach from the middle region MR to the corner portions may be reduced, to thereby prevent the non-conductive film from being unfilled in the corner portions.

[0128] Similarly, the fourth semiconductor chip **200**_d_ may be attached onto the third semiconductor chip **200**_c_ using a fourth adhesive layer **300**_d_. The fourth semiconductor chip **200**_d_ may be arranged such that a first surface **212**_d_ of a substrate **210**_d_ of the fourth semiconductor chip **200**_d_ faces the third semiconductor chip **200**_c_. The fourth semiconductor chip **200**_d_ may be stacked on the third semiconductor chip **200**_c_ via fourth conductive bumps. The fourth semiconductor chip **200**_d_ may be attached on the third semiconductor chip **200**_c_ by performing a thermal compression process. The fourth conductive bumps of the fourth semiconductor chip **200**_d_ may be bonded to second bonding pads **280**_c_ of the third semiconductor chip **200**_c_ through the thermal compression process. A first bonding pad **230**_d_ of the fourth semiconductor chip **200**_d_ may be electrically connected to the second bonding pad **280**_d_ of the third semiconductor chip **200**_d_ through the fourth conductive bump. The fourth adhesive layer **300**_d_ may include a non-conductive film (NCF).

[0129] In the thermal compression process, the non-conductive film may be liquefied and may have fluidity, and may flow between the fourth semiconductor chip **200**_d_ and the third semiconductor chip **200**_c_. The non-conductive film having fluidity may flow between the fourth conductive bumps and then may be cured to fill a space between the fourth conductive bumps. A portion of the cured fourth adhesive layer **300**_d_ may protrude from a side surface of the fourth semiconductor chip **200**_d_.

[0130] Second bump structures **260**_d_ having an oval shape may be disposed in diagonal regions that extend from a middle region MR of the fourth semiconductor chip **200**_d_ to four corner portions, respectively. The second bump structure **260**_d_ may be arranged such that the long axis Xa of the oval shape extends towards the middle region MR. A ratio (L/D) of the long axis to the short

axis of the second bump structure **260***d* may be at least 1.5.
[0131] When the non-conductive film having fluidity flows through the second bump structures **260***d* along a diagonal direction in the diagonal regions that have relatively long lengths, since the flow resistance coefficient CD of the second bump structure **260***d* is relatively low compared to the first bump structure, a flow rate of the non-conductive film flowing through the diagonal region may be relatively increased. Accordingly, the time for the non-conductive film to reach from the middle region MR to the corner portions may be reduced, to thereby prevent the non-conductive film from being unfilled in the corner portions.
[0132] Referring to FIG. **25**, a molding member **400** may be formed on the first wafer W**1** to cover side surfaces of the first, second, third and fourth semiconductor chips **200***a*, **200***b*, **200***c* and **200***d*.
[0133] In example embodiments, the molding member **400** may be formed to fill gaps between the first, second, third and fourth semiconductor chips **200***a*, **200***b*, **200***c* and **200***d*. The molding member **400** may expose an upper surface of the fourth semiconductor chip **200***d*. The molding member **400** may be formed using a polymer material such as an epoxy molding compound (EMC).
[0134] Then, the first wafer W**1** may be cut along the scribe lane region SA to form a buffer die **100** and the molding member **400** may be cut together to complete the semiconductor package **50** of FIG. **2**.
[0135] FIG. **26** is a graph showing a change in flow resistance coefficient according to a ratio of long and short axes of a non-conductive film around an oval conductive bump.
[0136] Referring to FIG. **26**, when a non-conductive film with fluidity flows in a direction parallel to the long axis direction of the oval conductive bump **260***a*, it can be seen that the flow resistance coefficient (CD) of the oval conductive bump **260***a* decreases as the long/short axis ratio (L/D) increases. Since a diagonal distance from the center of the chip to the corner of the chip is 1.43 times greater than a longitudinal distance to the edge of the chip, when the flow resistance coefficient (CD) of each of the conductive bumps **260***a* that are diagonally arranged is less than 0.7, the flow velocity may be 1.43 times faster, allowing the non-conductive film to reach the corner of the chip within the same time. When the long/short axis ratio (L/D) of the oval conductive bump **260***a* is at least 1.5, it may be possible to prevent the corner portion from being unfilled with the non-conductive film.
[0137] The semiconductor package may include semiconductor devices such as logic devices or memory devices. The semiconductor package may include logic devices such as central processing units (CPUs), main processing units (MPUs), or application processors (APs), or the like, and volatile memory devices such as DRAM devices, HBM devices, or non-volatile memory devices such as flash memory devices, PRAM devices, MRAM devices, ReRAM devices, or the like.
[0138] The foregoing is illustrative of example embodiments and is not to be construed as limiting thereof. Although a few example embodiments have been described, those skilled in the art will readily appreciate that many modifications are possible in example embodiments without materially departing from the novel teachings and advantages of the present invention. Accordingly, all such modifications are intended to be included within the scope of example embodiments as defined in the claims.

Claims

1. A semiconductor package, comprising: a buffer die; a plurality of core dies sequentially stacked on the buffer die via conductive bumps; and a plurality of adhesive layers between the plurality of core dies, the plurality of adhesive layers filling spaces between the conductive bumps and interconnecting the plurality of core dies; wherein each of the core dies of the plurality of core dies comprises a middle region and diagonal regions that extend from the middle region to four corner portions of a respective core die, and wherein the conductive bumps comprises: a plurality of first bump structures in the middle region, each first bump structure having a circular shape; and a plurality of second bump structures in each of the diagonal regions, each second bump structure having an elliptical shape.

2. The semiconductor package of claim 1, wherein the plurality of second bump structures are arranged such that longitudinal axes of the plurality of second bump structures extend toward the middle region.

3. The semiconductor package of claim 1, wherein a ratio of a major axis to a minor axis of each second bump structure of the plurality of second bump structures is at least 1.5, and wherein the major axis has a first length and the minor axis has a second length shorter than the first length.

4. The semiconductor package of claim 1, wherein each second bump structure of the plurality of second bump structures has a major axis length within a range of 15 μm to 40 μm and a minor axis length within a range of 8 μm to 20 μm, and wherein the minor axis length is shorter than the major axis length.

5. The semiconductor package of claim 1, wherein each first bump structure of the plurality of first bump structures has a diameter within a range of 10 μm to 20 μm.

6. The semiconductor package of claim 1, wherein each first bump structure of the plurality of first bump structures comprises (i) a first pillar bump having a circular shape on a first bonding pad in the middle region and (ii) a first solder bump on the first pillar bump, and wherein each second bump structure of the plurality of second bump structures comprises (i) a second pillar bump having an oval shape on a first bonding pad in each of the diagonal regions and (ii) a second solder bump on the second pillar bump.

7. The semiconductor package of claim 6, wherein the first pillar bump of each of the plurality of first bump structures and the second pillar bump of the each of the plurality of second bump structures comprise copper.

8. The semiconductor package of claim 1, wherein a length in an extension direction of each of the diagonal regions from the middle region is within a range of 4 mm to 6.5 mm.

9. The semiconductor package of claim 1, wherein each of the core dies of the plurality of core dies comprises (i) a substrate having a first surface and a second surface opposite to the first surface, and (ii) first bonding pads on the first surface of the substrate.

10. The semiconductor package of claim 1, wherein the plurality of adhesive layers comprise a non-conductive film (NCF).

11. A semiconductor package, comprising: a first semiconductor chip; a second semiconductor chip; a third semiconductor chip; a fourth semiconductor chip; and a plurality of adhesive layers, wherein the first semiconductor chip, the second semiconductor chip, the third semiconductor chip, and the fourth semiconductor chip are sequentially stacked using conductive bumps, wherein the plurality of adhesive layers are between the first semiconductor chip, the second semiconductor chip, the third semiconductor chip, and the fourth semiconductor chip, the plurality of adhesive layers filling spaces between the conductive bumps and interconnecting the first semiconductor chip, the second semiconductor chip, the third semiconductor chip, and the fourth semiconductor chip, wherein each of the first semiconductor chip, the second semiconductor chip, the third semiconductor chip, and the fourth semiconductor chip comprises (i) a substrate having a first surface and a second surface opposite to the first surface, and (ii) first bonding pads provided on the first surface of the substrate, wherein each of the first semiconductor chip, the second semiconductor chip, the third semiconductor chip, and the fourth semiconductor chip has a middle region and diagonal regions that extend from the middle region to four corner portions of a respective semiconductor chip, and wherein the conductive bumps comprises: a plurality of first bump structures on the first bonding pads in the middle region, each first bump structure having a circular shape, and a plurality of second bump structures on the first bonding pads in each of the diagonal regions, each second bump structure having an elliptical shape.

12. The semiconductor package of claim 11, wherein each of the first semiconductor chip, the second semiconductor chip, the third semiconductor chip, and the fourth semiconductor chip comprises through electrodes that penetrate the substrate of a respective semiconductor chip.

13. The semiconductor package of claim 11, wherein the plurality of second bump structures are arranged such that longitudinal axes of the plurality of second bump structures extend toward the middle region.

14. The semiconductor package of claim 11, wherein a ratio of a major axis to a minor axis of each second bump structure of the plurality of second bump structures is at least 1.5, and wherein the major axis has a first length and the minor axis has a second length shorter than the first length.

15. The semiconductor package of claim 11, wherein each of the plurality of second bump structures has a major axis length within a range of 15 μm to 40 μm and a minor axis length within a range of 8 μm to 20 μm, and wherein the minor axis length is shorter than the major axis length.

16. The semiconductor package of claim 11, wherein each of the plurality of first bump structures has a diameter within a range of 10 μm to 20 μm.

17. The semiconductor package of claim 11, wherein each of the plurality of first bump structures comprises (i) a first pillar bump having a circular shape on the first bonding pad in the middle region and (ii) a first solder bump on the first pillar bump, and wherein each of the plurality of second bump structures comprises a second pillar bump having an oval shape on the first bonding pad in each of the diagonal regions and (ii) a second solder bump on the second pillar bump.

18. The semiconductor package of claim 17, wherein the first pillar bump of each of the plurality of first bump structures and the second pillar bump of each of the plurality of second bump structures comprise copper.

19. The semiconductor package of claim 11, wherein the plurality of adhesive layers comprises a non-conductive film (NCF).

20. A semiconductor package, comprising: a first semiconductor chip comprising (i) a first substrate having a first surface and a second surface opposite the first surface, (ii) a plurality of first through electrodes penetrating the first substrate, and (iii) a plurality of first bonding pads provided on the second surface and electrically connected to the plurality of first through electrodes; a second semiconductor chip stacked on the second surface of the first semiconductor chip, the second semiconductor chip comprising (i) a second substrate having a third surface that faces the second surface and a fourth surface opposite to the third surface, and (ii) a plurality of second bonding pads on the third surface to respectively correspond to the plurality of first bonding pads; a plurality of conductive bumps interposed between the plurality of first bonding pads and the corresponding plurality of second bonding pads; and an adhesive layer between the first semiconductor chip and the second semiconductor chip, the adhesive layer filling spaces between the conductive bumps and interconnecting the first semiconductor chip and the second semiconductor chip, wherein the second substrate of the second semiconductor chip has diagonal regions that extend from a middle region to four corner portions of the second semiconductor chip, wherein the plurality of conductive bumps comprise bump structures on the plurality of second bonding pads in each of the diagonal regions, each bump structure having an elliptical shape, and wherein the bump structures are arranged such that longitudinal axes of the bump structures extend toward the middle region.